

NITROGEN AND ENERGY METABOLISM DURING THE EARLY ONTOGENY OF DIPLOID AND TRIPLOID RAINBOW TROUT (*SALMO GAIARDNERI* R.)

A. OLIVA-TELES and S. J. KAUSHIK*

Instituto de Zoologia, Faculdade de Ciências do Porto Pç Gomes Teixeira, 4000 Porto, Portugal and

*Laboratoire de Nutrition des Poissons, I.N.R.A., Saint Pée-sur-Nivelle, 64310 Ascaïn, France.

Telephone: (59)54-1054

(Received 26 June 1986)

Abstract—1. Patterns of nitrogen excretion and of oxygen consumption, and utilization of protein and nitrogen during the early development of diploid and triploid rainbow trout were measured.

2. Ammonia excretion ($\mu\text{g N} \cdot \text{ind}^{-1} \cdot \text{hr}^{-1}$) increased from 0.08 at the beginning to 0.25 at hatching and to 1.5 by the end of yolk absorption.

3. At similar stages, oxygen consumption rates were respectively 1, 4 and $30 \mu\text{g} \cdot \text{ind}^{-1} \cdot \text{hr}^{-1}$.

4. While fresh weight, water and protein contents remained constant during egg development, protein and energy utilization increased after hatching.

5. Protein was the major energy source during embryonic development, the utilization of lipids becoming predominant after hatching.

6 No significant differences were observed between diploids and triploids in their metabolic utilization of endogenous reserves.

I. INTRODUCTION

In commercial trout farming sexual maturation is considered a detrimental trait in terms of production because it reduces fish growth, survival and flesh quality. The obtention of sterile triploid fish has been investigated in several species as a genetical approach to prevent or at least reduce gonadal development and consequently avoid the negative traits related to it (Thorgaard, 1983; Chevassus and Blanc, 1983).

In salmonids, which reach sexual maturation at the end of the second year of growth, triploid production has been considered as a promising technique for the obtention of industrial numbers of big sterile fish. However the small amount of data available on performance comparisons of diploid–triploid rainbow trout (Lincoln and Scott, 1983; Thorgaard *et al.*, 1982) and our own experience, consistently shows that at least during the first life stage, mortality of triploid fishes is higher than that of diploid fishes.

As during their early ontogeny teleost fish must only rely on the endogenous reserves, we thought it would be important to study the metabolic utilization of body stores during this period in order to find any physiological explanation for the eventual differences between diploids and triploids. Our approach was based on studies of oxygen consumption and nitrogen excretion as well as modifications of body components during early development, from fertilization until yolk absorption.

II. MATERIAL AND METHODS

Diploid and triploid eggs were obtained from the same group of fishes. Triploid eggs were produced by a heat shock

technique (Chevassus *et al.*, 1983) at 28°C for 20 min. Eggs were kept at our facilities in a recirculated system under controlled temperature conditions ($12 \pm 0.1^\circ\text{C}$). Fertilization took place on 10 January, 1984 (day 1). Eyed stage was attained at day 13, and hatching took place at day 25–27. At the end of the trial (day 44) yolk was almost completely absorbed (total absorption at day 46). Triploidism was confirmed by chromosome counting: 90% of the examined eggs were triploid and 10% were diploid.

The metabolic measurements were made on replicate groups on different days from fertilization until day 44. Oxygen consumption was determined on a 8 ml respirometer with a YSI-microrespirometer. Triplicate groups of 10 eggs and duplicate groups of 2 eleutheroembryo were analysed each measurement day. The duration of measurement was such as to have a 20% decrease in oxygen saturation. For nitrogen (ammonia) excretion measurements triplicate groups of 15 eggs or duplicate groups of 5 eleutheroembryo were transferred at each measurement day to covered glass vials of 100 ml capacity where they were left for 5 hr. A control group with no fish was also prepared. After this time water was collected for ammonia measurement by the Nessler method. During the trial, ammonia never raised above $0.5 \text{ mg} \cdot \text{l}^{-1}$ which is considered far from the toxic level for rainbow trout eggs and eleutheroembryo.

Fresh weight, water, protein and energy content were also analysed during this period. Water was determined after drying at 104°C for 24 hr, protein ($N \times 6.25$) by the micro-Kjeldahl method (acid digestion) and energy with a Phillipson microbomb calorimeter.

Statistical analyses were performed according to the usual procedures given in Sokal and Rohlf (1981). Results are presented as mean $\pm$ SD.

III. RESULTS

As it can be seen in Fig. 1 oxygen consumption rates slowly rose before hatching. After hatching it increased 5–6 times and then more gradually until

*Send reprint requests to S. J. Kaushik.

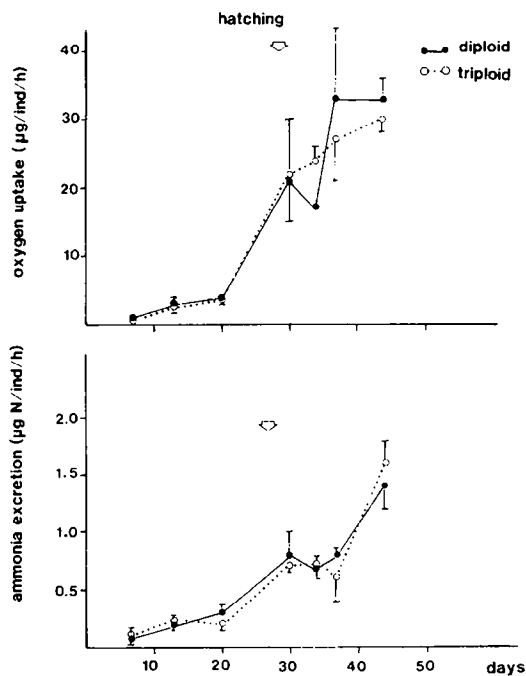


Fig. 1. Nitrogen excretion and oxygen consumption of diploid and triploid eggs and eleutheroembryo of rainbow trout.

yolk absorption. Nitrogen (ammonia) excretion rates also slowly rose before hatching. At hatching it increased 3 times, slowly declined thereafter for a few days and finally increased again at yolk absorption. No statistically significant differences were noticed in nitrogen excretion and oxygen consumption between diploid and triploid individuals at any sampling period except on day 20 where a significant reduction in oxygen uptake in diploid eleutheroembryo occurred (t -test, $P = 0.05$).

Table 3. Ammonia quotient values (A.Q.) during the early development of diploid and triploid rainbow trout

Day	Diploids	Triploids
7	0.21	0.25
13	0.20	0.23
20	0.17	0.17
30	0.12	0.09
37	0.07	0.07
44	0.12	0.16

Evolution of body composition of eggs and eleutheroembryo is presented in Table 1. All parameters were almost constant during the egg stages; after hatching fresh weight slightly decreased and then slowly increased until the end of yolk absorption. Water content continuously increased and protein and energy content decreased during this period.

Cumulative data on nitrogen and energy expenditure of the experimental animals is presented in Table 2, and the ammonia quotient (A.Q.) values (ratio of N:O on a molar basis) observed during early development are presented in Table 3. A correlation matrix of all parameters considered in this study is presented in Table 4. As only minor differences were observed between diploid and triploid individuals combined data were used in the calculations.

IV. DISCUSSION

Although a decrease in weight and changes in chemical composition during the egg development should be expected due to metabolic utilization of energy reserves, such a variation is so small that it probably followed within experimental error. Such a lack of variation in the chemical composition of the eggs was also noticed in salmonids and in other fishes (Hayes, 1949; Takahashi *et al.*, 1978; Kamler, 1976; Cetta and Capuzzo, 1982; Kimata, 1982, 1983; Dabrowski and Luczynski, 1984).

Table 1. Chemical composition of eggs and eleutheroembryo of diploid and triploid rainbow trout during development*

Ploidy	Component	Day					
		7	13	20	30	37	44
2n	Fresh wt (mg)	93.41 $\pm$ 4.31	98.95 $\pm$ 1.13†	99.53 $\pm$ 1.78	79.88 $\pm$ 7.41	112.41 $\pm$ 3.12	105.70 $\pm$ 4.64
	Water content (%)	66.53 $\pm$ 0.15	67.30 $\pm$ 2.08	57.93 $\pm$ 4.45	72.50 $\pm$ 0.26	72.85 $\pm$ 1.48	79.90 $\pm$ 0.95
	Protein (% DW)	68.70 $\pm$ 0.16†	70.86 $\pm$ 1.60	71.10 $\pm$ 0.26	62.45 $\pm$ 0.51†	66.56 $\pm$ 0.59	65.07 $\pm$ 1.22
	Energy (Kcal/g.DW)	5.67 $\pm$ 0.19	5.95 $\pm$ 0.38	5.88 $\pm$ 0.57	5.78 $\pm$ 0.11	5.53 $\pm$ 0.10	5.49 $\pm$ 0.21
3n	Fresh wt (mg)	95.19 $\pm$ 2.63	92.52 $\pm$ 2.37†	99.30 $\pm$ 2.18	87.30 $\pm$ 6.49	105.91 $\pm$ 10.11	102.30 $\pm$ 3.22
	Water content (%)	66.37 $\pm$ 0.60	65.73 $\pm$ 0.32	63.33 $\pm$ 1.19	69.40 $\pm$ 2.80	76.30 $\pm$ 7.21	78.87 $\pm$ 0.25
	Protein (% DW)	70.23 $\pm$ 0.71†	70.79 $\pm$ 1.36	71.21 $\pm$ 0.10	68.16 $\pm$ 0.56†	67.82 $\pm$ 0.71	66.84 $\pm$ 0.41
	Energy (Kcal/g.DW)	5.72 $\pm$ 0.14	5.68 $\pm$ 0.10	5.63 $\pm$ 0.29	5.68 $\pm$ 0.18	5.58 $\pm$ 0.34	5.60 $\pm$ 0.17

*Mean values $\pm$ SD.

†Values differing significantly on the same day ($P < 0.05$).

Table 2. Cumulative data on nitrogen and energy metabolism during the development of diploid and triploid rainbow trout

	Period	Before hatching		Hatching		Yolk absorption	
	Days	0-20		20-30		30-44	
	Ploidy	2n	3n	2n	3n	2n	3n
N excretion ($\mu\text{g N.NH}_4\text{.ind}^{-1}$)		64.0	60.0	132.5	106.4	240.4	303.5
O ₂ consumption ($\mu\text{g.ind}^{-1}$)		928.8	784.8	3000.0	3048.0	9126.0	8702.4
Energy expenditure (J.ind ⁻¹)		11.9	10.0	39.5	40.5	122.2	115.6
L:P (Energy basis)		0.26	0.18	1.02	1.58	2.44	1.57

Table 4. Correlation values of the parameters considered in this study

	1	2	3	4	5	6	
Age	1						
Fresh wt	2	0.466					
Water content	3	0.810	0.290				
Protein	4	-0.646	0.124	-0.906			
Energy content	5	-0.647	0.075	-0.925	0.986		
N excretion	6	0.913	0.315	-0.848	-0.745	-0.745	
O ₂ consumption	7	0.953	0.421	0.868	-0.726	-0.733	0.896

Correlation values above 0.640 are significant at 1% levels.

The decrease in fresh weight after hatching is due to the rejection of the chorion at hatching and its posterior increase is mostly explained by an increase in water content in the unfed eleutheroembryo. Protein and energy content continuously decreased after hatching and until the end of trial, reflecting the more intensive metabolism of the eleutheroembryo. Such a trend was also noticed by several authors in salmonids (Hayes, 1949; Rice and Stokes, 1974; Zeiton *et al.*, 1977) as well as in other fishes (Kamler, 1976; Davenport and Lonning, 1980).

Oxygen consumption and nitrogen excretion had been measured during the early development of salmonids (Hayes *et al.*, 1951; Bailey *et al.*, 1980; Kamler and Kato, 1983) as well as in other fishes (Kamler, 1976; Davenport and Lonning, 1980; Cetta and Capuzzo, 1982; Eldridge *et al.*, 1982; Kaushik *et al.*, 1982; Davenport, 1983; Dabrowski *et al.*, 1984).

Comparison of oxygen consumption between diploid and triploid fish was only done, to our knowledge, by Swarup (1959) in *Gasterosteus aculeatus* and by Benfey and Sutterlin (1984) in landlocked *Salmo salar*. Both authors found no differences between diploid and triploid animals in a unit body weight basis. However, Swarup (1959) concluded that on a unit area of body surface oxygen consumption of diploid fish was significantly superior to that of triploid fish. In our study average oxygen consumption of diploid animals was almost always slightly higher than in triploid fish, however neither in fresh weight basis nor in surface area basis can this difference be considered significant.

Knowledge of the nitrogen excretion rates during the early development of fish is very scarce. In salmonids such studies were only done in *Salmo gairdneri* (Smith, 1947; Rice and Stokes, 1974) and *Onchorhynchus gorbuscha* (Bailey *et al.*, 1980).

In this study nitrogen excretion was measured under the form of ammonia excretion, this possibly being not the only nitrogen excretion product. Rice and Stokes (1974) could not detect urea as a waste product in rainbow trout eggs and eleutheroembryo but stated that this could be caused by injury of the eggs. Urea formation in the early stages of development via the urea cycle pathways was detected in rainbow trout by Dépêche *et al.* (1979) but after hatching no more urea was detected. Yarzhombeck and Maszenikova (1971) detected ammonia, urea, ornithine and other nitrogen waste products in the eggs of rainbow trout and sea trout as well as in other fishes, ammonia accounting to only 46% (on a molar basis) of the excretory products of rainbow trout at the end of its embryonic development. In sea trout ammonia was found to increase from 31 to 53% of the excretory products (on a molar basis) during egg development. In *Pseudopleuronectes americanus* Ce-

tta and Capuzzo (1982) found that ammonia was the most important nitrogen waste product in the eggs while in young larvae nitrogen products other than ammonia and primary amine represented more than 50% of the total products excreted.

During this trial nitrogen excretion patterns were similar to those observed by Smith (1974), Rice and Stokes (1974) and Bailey *et al.* (1980). These two later authors did not refer, however, to a slight reduction in the ammonia excretion levels following the important excretion rates after hatching. Such a reduction can be detected from the results of Smith (1974) and it was also noticed in other fish species (Kaushik *et al.*, 1982; Dabrowski *et al.*, 1984; Davenport *et al.*, 1983).

Evolution of nitrogen excretion and oxygen consumption patterns during egg development were very similar. After hatching and until yolk absorption oxygen consumption continuously increased while nitrogen excretion first slightly decreased and finally considerably rose by the time of complete yolk absorption. Based on the theoretical maximum of 0.33 for complete aerobic protein degradation (Kutty, 1978) and on the assumption that ammonia is the main nitrogen waste product in rainbow trout, A.Q. values indicate a diminution of the importance of protein as an energy source during development, except at the end of the trial, where an increase in the A.Q. indicates an increase of protein utilization for metabolism possibly due to a degradation of body protein in the starving eleutheroembryo.

The energy expenditure during development showed to be equivalent in both diploid and triploid trout. The ratio of lipid energy to protein energy used for metabolism shows that protein was the most important energy source before hatching while thereafter lipids became more intensively used. These results do not agree with those of Hayes (1949) who suggested that salmonid eggs used fat as the primary source of energy while protein was predominant after hatching. A more important use of protein energy for metabolism during egg development and a predominant utilization of lipids after hatching was also noticed in other fish species by Takahashi *et al.* (1978), Kaushik *et al.* (1982), Davenport *et al.* (1983) and Dabrowski *et al.* (1984).

This study does not show the existence of any significative biochemical or metabolic difference between diploid and triploid rainbow trout during their early ontogeny but these results deserve further confirmation at later stages.

REFERENCES

- Bailey J. E., Rice S. D., Pella J. J. and Taylor S. G. (1980) Effects of seeding density of pink salmon *Oncorhynchus gorbuscha*, eggs on water chemistry, fry characteristics

- and fry survival in gravel incubators. *Fish. Bull.* **78**, 649–658.
- Benfey T. J. and Sutterlin A. M. (1984) Oxygen utilization by triploid landlocked Atlantic salmon (*Salmo salar* L.). *Aquaculture* **42**, 69–73.
- Cetta C. M. and Capuzzo J. M. (1982) Physiological aspects of embryonic and larval development of the winter flounder *Pseudopleuronectes americanus*. *Mar. Biol.* **71**, 327–337.
- Chevassus B. and Blanc J. M. (1983) Les recherches françaises en génétique des poissons. *Travaux Français de Limnologie*, pp. 105–109. Association Française de Limnologie (Ed.).
- Chevassus B., Quillet, G. et Chourrout D. (1983) Note technique: obtention d'animaux triploïdes chez la truite arc-en-ciel. *Bull. Fr. Piscic.* **290**, 161–164.
- Dabrowski K., Kaushik S. J. and Luquet P. (1984) Metabolic utilization of body stores during the early life of white fish, *Coregonus lavaretus* L. *J. Fish Biol.* **24**, 721–729.
- Dabrowski K. and Luczynski M. (1984) Utilization of body stores in embryonated ova and larvae of two coregonid species (*Coregonus lavaretus* L. and *C. albula* L.). *Comp. Biochem. Physiol.* **79A**, 329–334.
- Davenport J. (1983) Oxygen and the developing eggs and larvae of the lumpfish, *Cyclopterus lumpus*. *J. Mar. Biol. Ass. U.K.* **63**, 633–640.
- Davenport J. and Lønning S. (1980) Oxygen uptake in the developing eggs and larvae of the cod, *Gadus morhua* L. *J. Fish Biol.* **16**, 249–256.
- Davenport J., Lønning S. and Kjørsvik E. (1983) Ammonia output by eggs and larvae of the lumpsucker, *Cyclopterus lumpus*, the cod, *Gadus morhua* and the plaice, *Pleuronectes platessa*. *J. Mar. Biol. Ass. U.K.* **63**, 713–723.
- Dépêche J., Gilles R., Daufresne S. and Chiapello H. (1979) Urea content and urea production via the ornithine-urea cycle pathway during the ontogenic development of two teleost fishes. *Comp. Biochem. Physiol.* **63A**, 51–56.
- Eldridge M. B., Whipple J. A. and Bowers M. J. (1982) Bioenergetics and growth of striped bass, *Morone saxatilis*, embryos and larvae. *Fish. Bull.* **80**, 461–474.
- Hayes F. R. (1949) The growth, general chemistry, and temperature relations of salmonid eggs. *Q. Rev. Biol.* **24**, 281–308.
- Hayes F. R., Wilmot I. R. and Livingstone D. A. (1951) The oxygen consumption of the salmon egg in relation to development and activity. *J. exp. Zool.* **116**, 377–395.
- Kamler F. (1976) Variability of respiration and body composition during early developmental stages of carp. *Pol. Arch. Hydrobiol.* **23**, 431–485.
- Kamler E. and Kato T. (1983) Efficiency of yolk utilization by *Salmo gairdneri* in relation to incubation temperature and egg size. *Pol. Arch. Hydrobiol.* **30**, 271–306.
- Kaushik S. J., Dabrowski K. and Luquet P. (1982) Patterns of nitrogen excretion and oxygen consumption during ontogenesis of common carp (*Cyprinus carpio*). *Can. J. Fish. Aquat. Sci.* **39**, 1095–1105.
- Kimata M. (1982) Changes of chemical composition during early development of egg and larve in the half beack *Hemiramphus sajor* (Temminck and Schlegel). *Bull. Jap. Soc. Fish.* **48**, 1683–1671.
- Kimata M. (1983) Changes of chemical composition during early development in the red sea bream *Chrysophrys major* (Temminck and Schlegel) egg and larva. *J. Fac. Mar. Sci. Technol. Tokai Univ.* **16**, 213–223.
- Kutty M. N. (1978) Ammonia quotient in sockeye salmon (*Oncorhynchus nerka*). *J. Fish. Res. Bd. Can.* **35**, 1003–1005.
- Lincoln R. F. and Scott A. P. (1983) Production of all-female triploid rainbow trout. *Aquaculture* **30**, 375–380.
- Rice S. D. and Stokes R. M. (1974) Metabolism of nitrogenous wastes in the eggs and alevins of rainbow trout, *Salmo gairdneri* Richardson. In *The Early Life History of Fish* (Edited by Blaxter J. H. S.), pp. 325–337. Springer, Berlin.
- Smith S. (1947) Studies in the development of the rainbow trout (*Salmo irideus*). I. The heat production and nitrogenous excretion. *J. exp. Biol.* **23**, 357–373.
- Sokal R. R. and Rohlf F. J. (1981) *Biometry*. Freeman & Co., San Francisco.
- Swarup H. (1959) The oxygen consumption of diploid and triploid *Gasterosteus aculeatus* (L.). *J. Genet.* **56**, 156–160.
- Takahashi K., Ito K. and Sato, R. (1978) Glyconeogenesis in chum salmon alevin. *Tohoku J. agric. Res.* **29**, 146–158.
- Thorgaard G. H. (1983) Chromosome set manipulation and sex control in fish. *Fish. Physiol.* **9B**, 406–434.
- Thorgaard G. H., Rabinovitch P. S., Shen M. W., Gall G. A., Propp J. and Utter F. M. (1982) Triploid rainbow trout identified by flow cytometry. *Aquaculture* **29**, 305–309.
- Yarzhombeck A. A. and Maslennikova N. V. (1971) Nitrogenous metabolites of the eggs and larvae of various fishes. *J. Ichthyol.* **11**, 276–281.
- Zeitoun I. H., Ullrey D. E., Bergen W. G. and Magee W. T. (1977) DNA, RNA, Protein and free amino acids during ontogenesis of rainbow trout (*Salmo gairdneri*). *J. Fish. Res. Bd. Can.* **34**, 83–88.